EMDR Intervention – Cameroon Analysis Notes

This document provides a step-by-step account of the statistical analysis conducted for the study Evaluating the Effectiveness of EMDR Therapy on PTSD Among ICU Healthcare Professionals (doctoral dissertation, Daystar University). Its purpose is to offer a transparent record of methodological decisions, analytical workflows, and the interpretive reasoning that connects statistical outputs to clinical meaning. Each section follows a phase of the analysis, stating the rationale behind every choice. Results are interpreted within the study's real-world context, highlighting what the numbers reveal about the mental health burden faced by ICU healthcare workers and the impact of the intervention. By documenting both what was done and why it was done, this record is intended to support reproducibility, informed critique, and a clear understanding of how the findings were built from the ground up. In doing so, the client could understand and justify the statistical decisions involved in the analysis and understand the implications of the findings on the study.

By

Ruth Ushindi

Socio-Demographic Profiling

1. Overview

The socio-demographic analysis served four purposes:

1. To describe the characteristics of the entire study sample (treatment and control combined).
2. To profile participants within each group (treatment and control).
3. To compare socio-demographic characteristics between groups to assess baseline equivalence — ensuring that any post-intervention differences are attributable to the intervention rather than pre-existing differences.
4. To explore associations between selected socio-demographic variables to identify potential relationships or confounding factors that could influence study outcomes.

Because all variables were categorical, descriptive and inferential analyses relied on frequency-based methods rather than measures of central tendency.

2. Tests Conducted

A. Frequencies Analysis

- Purpose: To obtain counts and percentages for each socio-demographic category across the entire sample.
- Method: *Frequencies* procedure in SPSS (Analyze → Descriptive Statistics → Frequencies).
- Why appropriate:
  - All socio-demographic variables were nominal or ordinal, meaning that measures such as means and standard deviations would not be meaningful.
  - Frequencies and percentages provide a direct representation of category distribution without assuming equal intervals or a normal distribution.
- Alternative methods considered:
  - Means and SDs — rejected because they are only appropriate for continuous variables (e.g., age as a number), while the study used age and similar variables in grouped categories.
  - Graphs only — rejected as a primary method because they are less concise for statistical reporting; tables were preferred for clarity.

Table 1

Socio-Demographic Characteristics of Participants (N = 134)

Characteristic	n	%
Gender		
Female	67	50.0
Male	67	50.0
Age Group		
Up to 25 years	1	0.7
26–35 years	77	57.5
36–45 years	56	41.8
Marital Status		

Divorced	7	5.2
Married	69	51.5
Single	52	38.8
Widowed	6	4.5
Role		
Nurse	88	65.7
Physician	20	14.9
Support staff	26	19.4
Working Hours		
Below average hours	47	35.1
Average hours	48	35.8
Above average hours	39	29.1
ICU Experience Length		
1–5 years	66	49.3
6–10 years	65	48.5
11–15 years	3	2.2

Note. Percentages are based on valid responses.

Interpretation

The sample comprised an equal distribution of male and female participants (50.0% each). Most respondents were aged 26–35 years (57.5%), followed by those aged 36–45 years (41.8%), with only one participant aged up to 25 years (0.7%). Over half were married (51.5%), with singles comprising 38.8%, divorced 5.2%, and widowed 4.5%. The majority were nurses (65.7%), followed by support staff (19.4%) and physicians (14.9%). Working hours were nearly evenly split between below average (35.1%), average (35.8%), and above average (29.1%). Nearly half of participants reported 1–5 years (49.3%) or 6–10 years (48.5%) of ICU experience, with only 2.2% having 11–15 years.

B. Crosstabs for Group-wise Profiles

- Purpose: To compare socio-demographic distributions between treatment and control groups for baseline equivalence.
- Method: *Crosstabs* with Group as the column variable and each socio-demographic variable as the row variable, reporting row percentages.
- Why appropriate:
    - Allows side-by-side comparison of group distributions without implying an underlying numeric continuum.
    - Particularly suited to categorical baseline checks.
- Alternative methods considered:
    - Independent-samples t-test — rejected because it assumes continuous data with normally distributed values.
    - Logistic regression — rejected at baseline stage because the objective was descriptive equivalence rather than predictive modeling.

C. Chi-square Tests of Association

- Purpose: To test whether distributions of socio-demographic variables differed significantly between:

1. Treatment vs Control groups (baseline equivalence).
2. Selected socio-demographic pairs for broader context and potential confounders.

Method: Pearson's Chi-square test in SPSS Crosstabs (Statistics → Chi-square). Fisher's Exact test was used for 2×2 tables or when >20% of cells had expected counts <5.

Pairs tested and rationale:

1. Gender × Marital Status – Marital support may influence coping with trauma.
2. Gender × Professional Role – Different roles may have unequal gender representation, influencing exposure to trauma.
3. Gender × Years of ICU Experience (categorical) – Experience differences could confound PTSD outcomes.
4. Professional Role × Years of ICU Experience – Experience may vary systematically by role, affecting burnout risk.
5. Hospital Site × Professional Role – Site-specific staffing patterns may influence workload distribution.
6. Hospital Site × Gender – Gender representation differences between sites could reflect systemic staffing patterns.
7. Marital Status × Years of ICU Experience – Longer-serving staff may differ in relationship status, influencing support systems.
8. Group (Treatment vs Control) × Each socio-demographic variable – Direct check for baseline equivalence before intervention.

Why appropriate:

- All variables were categorical, satisfying Chi-square assumptions (with Fisher's Exact applied where small cell counts occurred).
- Chi-square tests provide statistical significance for associations without assuming interval-level data.

Alternative methods considered:

- Likelihood Ratio Chi-square – similar but Pearson's is standard for socio-demographic association testing unless sample sizes are very small.
- Cramér's V – useful for effect size, but secondary to establishing significance.

Table 2

Baseline Socio-Demographic Characteristics by Study Group (N = 134)

Characteristic	Control n (%)	Treatment n (%)	Total n (%)	χ^2(df)	p	Cramér's V
Gender						
Female	32 (47.8)	35 (52.2)	67 (50.0)	0.27(1)	.604	.045
Male	35 (52.2)	32 (47.8)	67 (50.0)			
Age group						
Up to 25 years	0 (0.0)	1 (1.5)	1 (0.7)	4.36(2)	.113	.181
26–35 years	44 (65.7)	33 (49.3)	77 (57.5)			
36–45 years	23 (34.3)	33 (49.3)	56 (41.8)			
Marital status						
Divorced	3 (4.5)	4 (6.0)	7 (5.2)	0.90(3)	.825	.082
Married	35 (52.2)	34 (50.7)	69 (51.5)			
Single	25 (37.3)	27 (40.3)	52 (38.8)			
Widowed	4 (6.0)	2 (3.0)	6 (4.5)			
Role						
Nurse	44 (65.7)	44 (65.7)	88 (65.7)	0.00(2)	1.000	.000
Physician	10 (14.9)	10 (14.9)	20 (14.9)			
Support staff	13 (19.4)	13 (19.4)	26 (19.4)			
ICU experience length						
1–5 years	30 (44.8)	36 (53.7)	66 (49.3)	4.79(2)	.091	.189
6–10 years	37 (55.2)	28 (41.8)	65 (48.5)			
11–15 years	0 (0.0)	3 (4.5)	3 (2.2)			
Working hours						
Below average	19 (28.4)	28 (41.8)	47 (35.1)	4.91(2)	.086	.192
Average	23 (34.3)	25 (37.3)	48 (35.8)			
Above average	25 (37.3)	14 (20.9)	39 (29.1)			

Note. Percentages are within groups. χ^2 = Pearson Chi-square test. Cramér's V is the effect size measure for categorical variables, interpreted as: small (.10), medium (.30), large (.50). p-values are 2-sided.

Interpretation

Baseline socio-demographic characteristics were balanced between treatment and control groups, with no statistically significant differences in gender, age group, marital status, role, ICU experience length, or working hours. Effect size estimates (Cramér's V) for all variables were small (< .20), supporting the absence of meaningful baseline imbalance. The largest effect sizes were observed for working hours (V = .192) and ICU experience length (V = .189), but these remained below the threshold for medium effects, suggesting any observed differences are unlikely to meaningfully bias subsequent outcome analyses.

Table 3

Association Between Key Sociodemographic Variables (N = 134)

Variable Pair	χ^2(df)	p	Cramér's V	Notes
Gender × Marital Status	4.27(3)	.234	.179	50% of cells < 5
Gender × Professional Role	7.84(2)	.020*	.242	Adequate counts
Gender × ICU Experience Length	3.45(2)	.179	.161	33% of cells < 5
Professional Role × ICU Experience Length	18.12(4)	.001**	.327	33% of cells < 5
Group × Professional Role	0.00(2)	1.000	.000	Adequate counts
Group × Gender	0.27(1)	.604	.045	Adequate counts
Marital Status × ICU Experience Length	3.11(6)	.795	.108	66.7% of cells < 5

Note. χ^2 = Pearson Chi-square test statistic. p-values are 2-tailed. Cramér's V interpreted as: small = .10, medium = .30, large = .50. Fisher's Exact test applied when >20% of cells had expected counts < 5 for 2×2 tables. *p < .05; **p < .01.

Interpretation:

Two statistically significant associations were found:

- Gender × Professional Role (p = .020, V = .242, small–medium effect) — suggesting certain roles had unequal gender representation.
- Professional Role × ICU Experience Length (p = .001, V = .327, medium effect) — indicating ICU experience distribution varied across roles.

These relationships could act as potential confounders in analyses involving gender, professional role, and ICU experience. All other tested associations were not statistically significant, with small effect sizes.

Overall Implications

1. **Sample Representativeness and Composition**
    - The sample was evenly split by gender and largely concentrated in the 26–45 year age range, representing a relatively young-to-mid-career workforce.
    - The high proportion of nurses (65.7%) reflects the staffing reality of ICU environments, where nurses are frontline providers.
    - Most participants reported 1–10 years of ICU experience, indicating a balance between early- and mid-career staff.
    - These distributions enhance the study's ecological validity, as they mirror common ICU workforce structures in similar contexts.
2. **Baseline Equivalence (Internal Validity)**
    - Treatment and control groups did not differ significantly on any socio-demographic characteristic.
    - Effect sizes were small, and none approached the medium threshold, supporting true baseline equivalence.
    - This strengthens causal inference: post-intervention differences are more likely attributable to the intervention rather than pre-existing group differences.
3. **Explored Associations (Potential Confounders)**
    - **Gender × Professional Role:** Significant, with a small-to-medium effect. This suggests role-specific exposure may be gendered (e.g., more female nurses, more male physicians), which could indirectly shape stress, burnout, or PTSD outcomes.
    - **Professional Role × ICU Experience Length:** Significant, with a medium effect. Different roles are associated with systematically different experience levels — for instance, physicians were more likely to fall into the longer-experience category.
    - These findings highlight the importance of controlling for role and experience (and sometimes gender) in downstream analyses, as they could confound intervention effects.
4. **Non-Significant Associations (Contextual Insight)**
    - Other socio-demographic associations (e.g., marital status, working hours) were not statistically significant and had small effect sizes.
    - This suggests that social support (via marital status) and workload (via reported hours) were relatively evenly distributed across groups and demographic categories.
    - While not confounders statistically, these variables remain clinically relevant and may still moderate intervention effects.
5. **Methodological Implications**
    - Using frequencies and percentages instead of means/SDs was methodologically sound since all variables were categorical.

- Reliance on Chi-square and effect size measures ensured appropriate, transparent assessment of group equivalence and inter-variable associations.
- Significant cross-demographic associations provide justification for including certain covariates in adjusted analyses (e.g., ANCOVA, logistic regression, or stratified analysis).

Summary:

The socio-demographic profiling confirms that treatment and control groups are balanced, strengthening the study's internal validity. At the same time, the significant associations (Gender × Role, Role × ICU Experience Length) identify plausible confounders, which you should statistically account for in outcome analyses. This dual finding — equivalence at baseline but internal demographic clustering — is common in clinical and occupational studies and underscores the value of careful covariate adjustment.

Objective 1: The prevalence of PTSD among ICU healthcare professionals in the selected general hospitals in Kumbo, Bui Division, Cameroon.

Step 1: Variable Creation

- Action: Created a binary categorical variable for PTSD status using the PCL-5 cut-off score (≥33 = probable PTSD, <33 = no PTSD).
- Purpose: Transform continuous PCL-5 scores into a format that directly answers the research question on prevalence (percentage of participants with probable PTSD).
- Appropriateness: A binary outcome aligns with epidemiological reporting standards for prevalence studies, making results interpretable for policy and clinical audiences.
- Other methods considered:
    - Using the continuous PCL-5 total score — rejected because prevalence requires a threshold to classify cases.
    - Creating severity categories — rejected because the stated objective does not address severity differences.

Step 2: Descriptive Statistics

- Test/Procedure: Frequencies and percentages (Analyze → Descriptive Statistics → Frequencies in SPSS GUI).
- Purpose: Quantify the proportion of healthcare professionals meeting the PTSD cut-off.
- Appropriateness: Frequencies and percentages are the standard descriptive approach for nominal variables when summarizing prevalence.
- Other methods considered:
    - Means and standard deviations — rejected because the variable is categorical, making central tendency measures inappropriate.

Prevalence of PTSD at Baseline

Table 4

Prevalence of Probable PTSD Among ICU Healthcare Professionals at Baseline

PTSD Status	n	%
PTSD unlikely	7	5.2
PTSD likely	127	94.8
Total	134	100.0

Note. Percentages are valid percentages. Probable PTSD was defined using the PCL-5 clinical cut-off score (≥33).

At baseline, a vast majority of ICU healthcare professionals met the criteria for probable PTSD. Out of 134 participants, 127 (94.8%) were classified as likely to have PTSD, while only 7 (5.2%) were classified as unlikely to have PTSD.

Step 3: Baseline Group Comparison

- Test: Chi-square test of independence (Analyze → Descriptive Statistics → Crosstabs in SPSS GUI).
- Purpose: Determine whether the proportion of probable PTSD cases differs significantly between the treatment and control groups at baseline.
- Appropriateness: Chi-square is suitable for testing associations between two categorical variables.
- Other methods considered:
    - Fisher's Exact Test — considered for small cell counts (<5), but if all cells have sufficient counts, Pearson's Chi-square is preferred for simplicity.
    - Independent-samples t-test — rejected because PTSD status is not continuous.

Baseline Equivalence in PTSD Status Between Groups

Table 5

PTSD Status by Group at Baseline

Group	PTSD unlikely, n (%)	PTSD likely, n (%)	Total, n (%)
Control	0 (0.0)	67 (100.0)	67 (100.0)
Treatment	7 (10.4)	60 (89.6)	67 (100.0)
Total	7 (5.2)	127 (94.8)	134 (100.0)

Note. Cramer's $V = .235$, $p = .007$. Percentages are valid percentages.

Interpretation and Implications

A chi-square test of independence was conducted to determine whether PTSD prevalence differed between the control and treatment groups at baseline. Results indicated a statistically significant difference in PTSD status between groups, $\chi^2(1, N = 134) = 7.39$, $p = .007$, Cramer's $V = .235$, suggesting a small-to-moderate effect size. Specifically, all participants in the control group were classified as likely to have PTSD (100%), whereas 10.4% of participants in the treatment group were classified as PTSD unlikely.

Importantly, the significant baseline difference in PTSD prevalence between the control and treatment groups suggests a potential confounding factor that should be accounted for in subsequent analyses, particularly when evaluating intervention effects. The observed effect size indicates that the association between group assignment and PTSD status at baseline was small-to-moderate, yet practically important given the high prevalence in both groups. This finding highlights the urgent need for targeted mental health interventions for ICU professionals and warrants careful consideration of baseline imbalances in PTSD status when interpreting outcome changes.

Objective 2: The influence of socio-demographic characteristics on the severity of PTSD symptoms among ICU healthcare professionals in selected general hospitals, Kumbo, Bui Division, Cameroon.

The dependent variable was the total PTSD score at baseline, measured using the PTSD Checklist for DSM-5 (PCL-5), which is a continuous scale (range: 0–80). Higher scores indicate greater PTSD severity. Six socio-demographic variables—age group, gender, marital status, professional role, ICU experience length, and working-hours category—were measured as categorical variables.

1) Descriptive stage (prior to modeling)

- Purpose: Describe the distribution of PTSD scores overall and by each (collapsed) socio-demographic group; check for obvious anomalies/outliers through descriptive statistics.
- Categories were collapsed for analysis to avoid small cell counts, while the original distribution is presented in the descriptive profile.
  This ensured stable estimates, met minimum cell sizes, and reduced assumption violations (e.g., heterogeneity of variance) given N=134.
- Why appropriate: Provides context for the inferential results; aligns with APA/biostat practice to present means/SDs (continuous outcome) before tests.
- Alternatives considered but not used: Reporting only prevalence above the PCL-5 cut-off (33) would ignore severity, which is this objective's focus.

Results

Table 6

Descriptive Statistics of PTSD Symptom Severity (PCL-5 Total Score) by Socio-Demographic Characteristics at Baseline

Variable	Category	*n*	Mean	SD	95% CI for Mean
Overall	—	134	42.22	5.55	[41.27, 43.17]
Age group	Up to 35 years	90	41.79	5.31	[40.60, 42.99]
	36+ years	44	42.82	5.88	[41.25, 44.39]
Marital status	Married	69	41.88	5.78	[40.49, 43.27]
	Unmarried	65	42.58	5.32	[41.27, 43.90]
ICU experience length	Up to 5 years	66	42.33	6.24	[40.80, 43.87]
	6+ years	68	42.12	4.84	[40.95, 43.29]
Gender	Female	67	41.97	5.66	[40.59, 43.35]
	Male	67	42.48	5.48	[41.14, 43.81]
Role	Nurse	88	42.19	5.68	[40.99, 43.40]
	Physician	20	40.85	5.41	[38.32, 43.38]
	Support staff	26	43.38	5.16	[41.30, 45.47]
Working hours	Below average	47	42.30	5.63	[40.64, 43.95]
	Average	48	42.08	5.57	[40.47, 43.70]
	Above average	39	42.31	5.58	[40.50, 44.12]

Note. PTSD symptom severity measured with the PCL-5. Higher scores indicate greater symptom severity. Categories reflect collapsed groups for inferential analysis; original distributions are reported in the socio-demographic profile. CI = confidence interval; SD = standard deviation.

Interpretation

At baseline, ICU healthcare professionals had a mean PTSD symptom severity score of 42.22 (*SD* = 5.55), well above the PCL-5 diagnostic cut-off score of 33. Across socio-demographic groups, mean scores were consistently above the clinical threshold, indicating high symptom burden regardless of age, marital status, ICU experience, gender, role, or working hours. Variations in mean scores between categories were modest, with the largest difference observed between support staff (*M* = 43.38) and physicians (*M* = 40.85). Confidence intervals for most subgroups overlapped substantially, suggesting limited unadjusted differences.

Implications

These findings indicate a pervasively high prevalence and severity of PTSD symptoms across the ICU workforce, suggesting that the risk is not confined to a particular socio-demographic subgroup. This implies that intervention strategies should be implemented universally across ICU roles and demographic categories, while also allowing for subgroup-specific adaptations if further inferential testing identifies significant predictors. The relatively small mean differences between groups may reflect the uniform exposure to high-stress ICU environments in this setting.

2) Main inferential model

Method used: General Linear Model (GLM) – Univariate (main-effects model)

- DV: Continuous PTSD total score.
- Fixed factors: All (collapsed) socio-demographics entered simultaneously (e.g., Gender, Age group, Marital status, Role, Working hours, ICU experience if retained after collapsing).
- Model: Main effects only (no interactions) to preserve power and avoid sparse cells.
- Multiple comparisons: Bonferroni pairwise tests for factors with >2 levels.
- Assumptions checked:
    - Levene's test (homogeneity of variances)
    - Residual plots / boxplots (to flag outliers and gross non-normality)
- Effect size: Partial eta-squared (ηp^2) reported for each factor.

Why appropriate:

- Handles a continuous outcome with multiple categorical predictors in one model, giving adjusted (confounder-controlled) tests for each factor.
- Simultaneous entry prevents misleading one-by-one results due to confounding (e.g., role vs. working hours overlap).

Methods considered but rejected (and why)

- Separate one-way ANOVAs / multiple t-tests per factor:
    - Rejected due to inflated Type I error and no adjustment for confounding between socio-demographics.

- Multiple Linear Regression with dummy-coded factors (entered stepwise):
    - Mathematically similar to GLM, but we prioritized GLM (Univariate) in the GUI to keep factor structure explicit, apply built-in post-hoc/least-squares means, and simplify APA reporting.
    - Stepwise selection adds data-driven bias; we preferred a theory-driven full main-effects model.
- Non-parametric tests (Mann–Whitney U / Kruskal–Wallis):
    - Rejected because GLM assumptions were reasonably met after collapsing categories; non-parametrics complicate multi-factor adjustment and effect-size reporting.
- Including interactions among factors:
    - Rejected to preserve power and avoid very small cells given sample size; interactions were beyond the scope of Objective 2.

Results

Table 7

Levene's Test of Equality of Error Variances for Baseline PTSD Score

Dependent Variable	Levene Statistic	df_1	df_2	*p*
PTSD-Baseline	2.60	37	58	.001

Note. Based on the mean. A significant result indicates heterogeneity of variances across groups.

Levene's test indicated heterogeneity of variances across groups, $F(37, 58) = 2.60$, $p = .001$. However, as none of the socio-demographic predictors reached statistical significance in the GLM, no robust follow-up tests (e.g., Welch's ANOVA) were pursued, as they would not change the conclusion.

Table 8

General Linear Model: Tests of Between-Subjects Effects for Baseline PTSD Score by Socio-Demographic Characteristics

Source	df	F	*p*	Partial η^2
Age group	1, 58	0.003	.960	.000
Marital status	1, 58	0.001	.971	.000
Role	2, 58	0.410	.666	.014
ICU experience	1, 58	0.201	.656	.003
Working hours	2, 58	0.259	.773	.009
Gender	1, 58	0.123	.727	.002
Corrected Model	75, 58	0.773	.854	.500

Note. Partial η^2 values represent the proportion of variance in baseline PTSD scores uniquely explained by each predictor after accounting for other variables in the model.

In the adjusted GLM, there were no main effects for any predictor (p values > .05), and all partial η^2 values were negligible (< .02), indicating no meaningful influence of these characteristics on baseline PTSD severity in this sample.

Interpretation

None of the socio-demographic variables were significantly associated with PTSD severity at baseline, and effect sizes were negligible, suggesting no meaningful influence of these characteristics on symptom severity in this sample.

Implications

The lack of statistically significant associations between socio-demographic characteristics and baseline PTSD severity in this study suggests that symptom levels were relatively uniform across demographic subgroups. This finding indicates that, within this population, factors such as age, gender, marital status, professional role, ICU experience length, and working hours may have limited influence on the initial severity of PTSD symptoms.

Practically, this suggests that interventions, prevention programs, and resource allocation targeting PTSD in similar settings may not need to be stratified primarily by these socio-demographic factors. Instead, attention may be better directed toward other determinants, such as clinical presentation, type or intensity of trauma exposure, and workplace or psychosocial support systems.

From a methodological standpoint, the absence of significant effects reduces concerns about confounding from these variables in subsequent analyses, supporting their optional rather than essential inclusion as covariates in future models. Nevertheless, given the significant Levene's test indicating heterogeneity of variances, results should be interpreted with caution. Additionally, the possibility of small but undetected effects cannot be ruled out due to sample size limitations.

Objective 3: Establish the comorbidities associated with PTSD among ICU healthcare professionals

Step 1: Variable preparation

- **Purpose:**
  - Convert continuous scale scores (HADS-D, HADS-A, and MBI subscales) into binary categorical variables representing *presence* vs *absence* of each comorbidity, following the pre-specified scoring thresholds.
- **Details:**
  - Anxiety: From HADS-A (items 8–14), total score ≥ 8 = potential anxiety.
  - Depression: From HADS-D (items 1–7), total score ≥ 8 = potential depression.
  - Burnout: Defined as high Emotional Exhaustion (≥ 27) AND either high Depersonalization (≥ 13) or low Personal Accomplishment (≤ 31).
- **Why this approach:**
  - Binary classification aligns with clinical screening thresholds.
  - Matches the operational definitions in the protocol, ensuring interpretability for clinical and policy audiences.

Step 2: Descriptive statistics

- **Purpose:** Provide an overview of the prevalence of anxiety, depression, and burnout in the sample.
- **Method used:**
  - **Frequencies and percentages**
- **Why appropriate:**
  - The variables are binary (nominal), making frequencies the clearest representation of prevalence.
- **Why not means/SDs:**
  - These are categorical classifications; means would be meaningless in this context. Percentages better convey the proportion affected.

Table 9

Prevalence of Depression, Anxiety, and Burnout Among ICU Healthcare Professionals at Baseline (N = 134)

Variable	Category	n	%
Depression	Absent	23	17.2
	Present	111	82.8
Anxiety	Absent	20	14.9
	Present	114	85.1
Burnout status	At risk	108	80.6
	Burnout	8	6.0
	No burnout	18	13.4

Note. Depression and anxiety classified using HADS ≥ 8 threshold; burnout classified according to MBI subscale criteria. Percentages are valid percentages.

Interpretation

At baseline, a substantial majority of ICU healthcare professionals met the screening criteria for both depression (82.8%) and anxiety (85.1%), indicating a high psychological symptom burden in this sample. Burnout classification revealed that 80.6% were at risk, with 6.0% meeting full burnout criteria, and only 13.4% showing no burnout indicators. These findings suggest that psychological distress and occupational strain were prevalent across the workforce prior to intervention.

Implications

- Clinical: The high prevalence of depression, anxiety, and burnout underscores the urgent need for targeted mental health support and preventive interventions in ICU settings.
- Organizational: These levels of distress can impair job performance, increase turnover, and impact patient care quality, suggesting a need for systemic workplace wellness programs.
- Research: The high baseline prevalence rates provide a reference point for evaluating the impact of subsequent interventions and for exploring comorbidity patterns with PTSD.

Step 3: Bivariate associations

- **Purpose:** Determine whether PTSD status is associated with each comorbidity.
- **Method used:**
  - **Chi-square test of independence** via *Analyze → Descriptive Statistics → Crosstabs* in SPSS.
  - PTSD status (≥ 33 = likely PTSD; < 33 = unlikely) as the dependent variable, comorbidity variables as independent variables.
  - Effect sizes reported (Phi coefficient for 2×2 tables).
- **Why appropriate:**
  - Both variables are categorical.
  - Chi-square is the standard non-parametric test for assessing association between nominal variables.
- **Handling small expected counts:**
  - When >20% of cells had an expected count < 5, Fisher's Exact Test was used (SPSS option within Crosstabs).
- **Why not other tests:**
  - Logistic regression (binary outcome) could also be used, but it was rejected at this stage to maintain simplicity and transparency in initial association testing. Regression was reserved for possible adjusted analyses later.

- t-tests or ANOVA were not suitable since PTSD status and comorbidities are both categorical, not continuous.

Table 10

Associations Between PTSD Status and Depression, Anxiety, and Burnout at Baseline (N = 134)

Variable	χ^2	df	p	Effect size (Cramer's V)
Depression status	1.53	1	.216	0.11
Anxiety status	1.08	1	.298	0.09
Burnout status	0.47	2	.791	0.06

Note. Depression and anxiety classified using HADS ≥ 8 threshold; burnout classified according to MBI subscale criteria. Effect sizes interpreted as per Cohen's guidelines: V = 0.06–0.11 (small).

Interpretation

No statistically significant associations were observed between PTSD status and depression, anxiety, or burnout at baseline (all $p > .05$). Effect sizes were small in magnitude, suggesting weak associations between PTSD and these comorbidities in this sample.

Although the majority of participants with PTSD also had depression, anxiety, or were at risk for burnout, these patterns did not differ significantly from those without PTSD. This may be partly due to the small number of participants classified as "PTSD unlikely" (n = 7), which limits statistical power to detect meaningful differences.

Implications

- Clinical: Even without statistically significant differences, the high co-occurrence rates (e.g., 93.7% of those with depression also had PTSD) underscore the importance of routine screening for multiple mental health concerns in ICU settings.
- Research: Future studies should recruit larger samples or use longitudinal designs to better detect possible interactions between PTSD and other forms of psychological distress.
- Workforce support: Given the overlap in symptomatology, workplace interventions should address PTSD, depression, anxiety, and burnout concurrently rather than in isolation.

Objective 4: Evaluate the effectiveness of EMDR therapy in treating PTSD among ICU healthcare professionals

Step 1 – Descriptive Statistics

Purpose:

- To summarise PCL-5 scores at baseline, midline, and endline for both treatment and control groups, giving an overview of central tendency and variability.
- To show raw score changes before adjusting for baseline differences.

Why this method was appropriate:

- Means and standard deviations are the standard descriptive summary for continuous outcome measures.
- This allowed a visual and numerical comparison of score trends over time between groups.

Other methods considered & rejected:

- Frequencies/percentages would not have been suitable here because the PCL-5 is a continuous scale, and summarising it categorically would lose valuable variation.
- Median/IQR could have been reported if the data were highly skewed, but inspection showed approximate normality.

Results

Table 11

Means and Standard Deviations of PTSD Scores by Group and Time Point

Time Point	Group	*n*	M	SD	95% CI (Lower, Upper)
Baseline	Control	51	44.20	3.85	43.11, 45.28
	Treatment	51	40.92	6.14	39.19, 42.65
Midline	Control	51	43.06	5.44	41.53, 44.59
	Treatment	51	19.33	5.76	17.71, 20.95
Endline	Control	51	43.24	6.36	41.45, 45.03
	Treatment	51	8.80	3.04	7.95, 9.66

Note. PTSD scores were assessed using the PCL-5, with higher scores indicating greater PTSD symptom severity.

Statistical reporting (APA style)

At baseline, mean PTSD scores were 44.20 (*SD* = 3.85) for the control group and 40.92 (*SD* = 6.14) for the treatment group. At midline, control scores remained relatively stable (M = 43.06, *SD* = 5.44), whereas treatment scores decreased markedly (M = 19.33, *SD* = 5.76). By endline, control scores were largely unchanged from baseline (M = 43.24, *SD* = 6.36), but treatment scores had further reduced to M = 8.80 (*SD* = 3.04).

Interpretation

The descriptive results suggest a substantial and sustained reduction in PTSD symptom severity in the treatment group following EMDR therapy, with scores decreasing well below the clinical cut-off for

probable PTSD by midline and continuing to decline at endline. In contrast, the control group's scores showed no meaningful change over the same period.

Implications

These findings provide preliminary descriptive evidence for the short- and long-term effectiveness of EMDR in reducing PTSD symptoms among ICU healthcare professionals. If confirmed by inferential analyses, such improvements could have meaningful clinical and occupational health implications, potentially enhancing staff resilience and reducing mental health burden in high-stress clinical settings.

Step 2 – General Linear Model (GLM) Repeated-Measures ANOVA

Purpose:

- To assess within-subject changes in PTSD severity over time (baseline → midline → endline) and between-group differences (treatment vs control), as well as any group × time interaction.

Why this method was appropriate:

- The same participants were measured at three time points, creating repeated measures data.
- Repeated-measures ANOVA accounts for within-subject correlations and allows testing of time, group, and interaction effects in one model.

Other methods considered & rejected:

- Multiple independent t-tests for each time point would increase the risk of Type I error and ignore within-subject correlations.
- Mixed models could have been used but would have been unnecessarily complex for the dataset size and structure.

Results

Table 12

Repeated Measures ANOVA for PTSD Scores

Effect	F	df	p	η^2_p
Time	459.91	2, 99	<.001	.903
Time × Group	402.66	2, 99	<.001	.891
Group (Between)	622.32	1, 100	<.001	.862

Note. df = degrees of freedom; η^2_p = partial eta squared.

- A repeated-measures ANOVA showed a statistically significant main effect of time, $F(2, 99) = 459.91$, $p < .001$, $\eta^2_p = .903$, indicating that PTSD scores changed significantly over the three time points.
- There was also a significant time × group interaction, $F(2, 99) = 402.66$, $p < .001$, $\eta^2_p = .891$, showing that the change over time differed between the treatment and control groups.

- The between-subjects effect for group was also significant, $F(1, 100) = 622.32$, $p < .001$, $\eta^2_p = .862$, with the treatment group having lower average PTSD scores across time.

Results

Table 13

Bonferroni-Adjusted Pairwise Comparisons for PTSD Scores Across Time

Time Comparison	Mean Difference	SE	95% CI (Lower, Upper)	p
Baseline – Midline	11.36	0.55	10.02, 12.71	<.001
Baseline – Endline	16.54	0.55	15.19, 17.89	<.001
Midline – Endline	5.18	0.55	3.84, 6.51	<.001

Note. Positive values indicate reductions in PTSD scores over time.

- Bonferroni-adjusted pairwise comparisons revealed significant reductions in PTSD scores from baseline to midline (M diff = 11.36, $p < .001$) and from baseline to endline (M diff = 16.54, $p < .001$).
- The reduction from midline to endline was also significant (M diff = 5.18, $p < .001$).

Step 3 – ANCOVA (Midline Analysis)

Purpose:

- To test for a short-term treatment effect at midline while adjusting for baseline PCL-5 scores.

Why this method was appropriate:

- ANCOVA removes baseline variability, providing a more accurate estimate of the treatment effect.
- Adjusting for baseline ensures differences at follow-up are not simply due to initial group imbalances.

Other methods considered & rejected:

- Independent samples t-test: less statistical power and no adjustment for baseline differences.
- Analysis of change scores: can be less efficient and sensitive to regression-to-the-mean effects.

Results

Table 14

Adjusted Means and Standard Errors of PTSD Scores at Midline by Group

Group	Adjusted Mean	SE	95% CI Lower	95% CI Upper
Control	42.32	0.66	41.02	43.63
Treatment	20.30	0.66	18.99	21.60

- An ANCOVA was conducted to compare PTSD scores between the treatment and control groups at midline, while controlling for baseline PTSD scores.
- The adjusted means indicate that, after controlling for baseline PTSD, the treatment group had a substantially lower PTSD score ($M = 20.30$, SE = 0.66) than the control group ($M = 42.32$, SE = 0.66).

Table 15

ANCOVA Summary for PTSD Scores at Midline (Baseline PTSD as Covariate)

Source	SS	df	MS	F	p	η^2_p
Baseline PTSD	817.31	1	817.31	35.14	<.001	.237
Group	12038.26	1	12038.26	517.57	<.001	.821
Error	2628.27	113	23.26			

- The ANCOVA revealed a statistically significant main effect of group, $F(1, 113) = 517.57$, $p < .001$, $\eta^2_p = .821$, indicating that the EMDR intervention was associated with markedly lower PTSD scores at midline compared to the control condition.
- Pairwise comparisons confirmed that the treatment group's adjusted mean was 22.03 points lower than that of the control group ($p < .001$), representing a clinically meaningful difference well above the 10-point reduction threshold for significant change on the PCL-5.

Interpretation

- The findings demonstrate that the treatment group experienced a large and statistically significant reduction in PTSD symptoms at midline compared to the control group, even after controlling for baseline differences.
- The effect size ($\eta^2_p = .821$) is exceptionally large, indicating that the intervention accounted for over 82% of the variance in midline PTSD scores after adjusting for baseline severity.
- From a clinical perspective, this effect represents not only statistical significance but also substantial clinical benefit, as the mean reduction far exceeded the minimum 10-point PCL-5 threshold for meaningful improvement.

Implications

- Rapid Symptom Reduction: EMDR produced substantial PTSD symptom relief after a short treatment period, suggesting it is a rapid-acting intervention suitable for high-stress occupational settings.
- High Clinical Relevance: The difference between groups was more than double the clinically meaningful change threshold, reinforcing EMDR's potential as a primary treatment option in trauma-exposed healthcare populations.
- Policy and Practice: Findings support the inclusion of EMDR in occupational health programs for ICU staff, especially in regions with limited access to mental health resources, as benefits appear even in the short term.

Step 4 – ANCOVA (Endline Analysis)

Purpose:

- To test for a long-term treatment effect at endline while controlling for baseline PCL-5 scores.

Why this method was appropriate:

- Same rationale as Step 3, but here the focus is on sustained effects at the 3-month follow-up.

Other methods considered & rejected:

- Same as for Step 3 — ANCOVA was preferred for its baseline adjustment and increased statistical power.

Results

Table 16

Adjusted Means and Standard Errors of PTSD Scores at Endline by Group

Group	Adjusted Mean	SE	95% CI Lower	95% CI Upper
Control	42.62	0.66	41.30	43.94
Treatment	9.42	0.66	8.10	10.74

- An ANCOVA was conducted to compare PTSD scores between the treatment and control groups at endline (3-month follow-up), while controlling for baseline PTSD scores.
- The adjusted means indicate that, after controlling for baseline PTSD, the treatment group had a markedly lower PTSD score (M = 9.42, SE = 0.66) compared to the control group (M = 42.62, SE = 0.66).

Table 17

ANCOVA Summary for PTSD Scores at Endline (Baseline PTSD as Covariate)

Source	SS	df	MS	F	p	η^2_p
Baseline PTSD	372.88	1	372.88	17.46	<.001	.150
Group	25454.86	1	25454.86	1191.88	<.001	.923
Error	2114.34	99	21.36			

- The ANCOVA revealed a statistically significant main effect of group, $F(1, 99) = 1191.88$, $p < .001$, $\eta^2_p = .923$, indicating that EMDR was associated with dramatically lower PTSD scores at the 3-month follow-up compared to the control condition.
- Pairwise comparisons confirmed that the treatment group's adjusted mean was 33.20 points lower than the control group ($p < .001$), representing a clinically massive difference well above the 10-point reduction threshold for meaningful change on the PCL-5.

Interpretation

- The results indicate that the treatment group maintained and even deepened their PTSD symptom reductions over the follow-up period. After controlling for baseline severity, the treatment effect size ($\eta^2_p = .923$) was extremely large, explaining more than 92% of the variance in endline PTSD scores.

- This demonstrates that the substantial benefits of EMDR observed at midline were sustained over time, with participants continuing to report PTSD symptom levels far below the clinical threshold.

Implications

- Sustained Treatment Gains: EMDR effects persisted over a 3-month follow-up, indicating long-term durability of symptom relief without apparent relapse.
- Clinical Significance: The difference between groups far exceeded the clinically meaningful threshold, suggesting EMDR achieves both immediate and lasting remission in PTSD among ICU healthcare workers.
- Workforce and Policy Impact: Sustained improvements could translate to better occupational functioning, reduced burnout risk, and lower healthcare costs associated with chronic PTSD.
- Scalability: The findings strengthen the case for integrating EMDR into mental health protocols for trauma-exposed professionals in resource-limited healthcare systems.

Step 5 – Remission Analysis (Frequencies & Crosstabs)

Purpose:

- To determine the proportion of participants in each group who achieved remission (≥10-point reduction in PCL-5 from baseline) at midline and at endline, and to compare these proportions statistically.

Why this method was appropriate:

- Crosstabulation with Chi-square or Fisher's Exact test is the standard approach for testing associations between two categorical variables (group × remission status).
- This analysis provides a clinically interpretable outcome alongside continuous score analysis.

Other methods considered & rejected:

- Logistic regression could have been used for adjusted remission comparisons, but the primary goal here was to present clear, unadjusted group differences for clinical interpretation.
- Reporting only mean score changes would not capture the clinically relevant outcome of remission.

Results

Table 18

PTSD Remission Status by Group and Time

Time Point	Group	Remission n (%)	No Remission n (%)
Midline	Treatment	56 (96.6%)	2 (3.4%)
	Control	0 (0.0%)	58 (100.0%)
Endline	Treatment	51 (100.0%)	0 (0.0%)
	Control	3 (5.9%)	48 (94.1%)

- Analysis of remission status (≥10-point reduction in PCL-5 scores) indicated dramatic differences between groups.
- At midline, 96.6% of the treatment group achieved remission, compared with none in the control group, $\chi^2(1, N = 116) = 108.27, p < .001$.
- At endline, remission remained high in the treatment group (100%), while only 5.9% of the control group achieved remission, $\chi^2(1, N = 102) = 90.67, p < .001$.

Overall Interpretation

- The results provide compelling evidence for the effectiveness of EMDR therapy in reducing PTSD symptoms among ICU healthcare professionals.
- Large effect sizes for both the main effect of time ($\eta^2_p = .903$) and the time × group interaction ($\eta^2_p = .891$) demonstrate that the intervention produced substantial improvements that were not observed in the control group.
- The remission data highlight the clinical relevance of these changes: nearly all participants in the treatment group achieved clinically significant symptom reduction by midline, and these gains were sustained or improved at endline.
- The lack of change in the control group underscores that the improvements were unlikely due to natural recovery or placebo effects, but rather to the EMDR intervention itself.

Implications

- Clinical Practice: EMDR therapy can produce rapid and sustained reductions in PTSD symptoms in high-stress healthcare environments, even after a relatively short intervention period.
- Workforce Well-being: Given the near-complete remission rates in the treatment group, EMDR could be a critical tool in protecting the mental health of ICU staff, potentially reducing burnout, absenteeism, and turnover.
- Implementation: Hospitals should consider integrating EMDR into staff support programs, especially in departments regularly exposed to traumatic events.
- Future Research: Studies should explore the impact of EMDR on other psychological outcomes, such as anxiety, depression, and burnout, and evaluate cost-effectiveness for broader health system adoption.

Integrated Interpretation

1. The repeated measures GLM revealed that PTSD symptoms changed significantly over time, with the intervention group demonstrating a markedly steeper decline than the control group.
2. At midline, after controlling for baseline PTSD scores, the treatment group's PTSD severity was on average 22 points lower than the control group, a difference representing both statistical and clinical significance.

3. At 3-month follow-up, this difference had widened to 33 points, indicating sustained and deepened treatment benefits over time.
4. Importantly, these statistical improvements translated into clinical recovery. By midline, 96.6% of participants in the treatment group achieved remission (≥10-point reduction on the PCL-5), compared with none in the control group, $\chi^2(1, N = 116) = 108.27$, $p < .001$. At endline, remission remained at 100% in the treatment group, while only 5.9% of the control group achieved remission, $\chi^2(1, N = 102) = 90.67$, $p < .001$.
5. Together, these analyses indicate that EMDR therapy not only produces rapid and substantial reductions in PTSD symptoms but also sustains these gains over time, with nearly universal remission among treated healthcare professionals. The convergence of strong statistical effects and near-total clinical recovery highlights EMDR's exceptional efficacy in this high-risk population.